

Bidirectional gRPC Streaming with HTTPS & H2C

Secure External + Efficient Internal Communication

- External: gRPC over **HTTPS** via Tyk
- Internal: gRPC over **H2C** (plain HTTP/2)
- Common in **Kubernetes** environments
 - TLS termination at ingress
 - Plain HTTP inside cluster

gRPC client ↔ Tyk Gateway (HTTPS) ↔ gRPC Server (H2C)

Prerequisites

- A gRPC server (e.g. `route_guide`)
 - Exposed at `localhost:10000`, TLS disabled
- A gRPC client (sample or your app)
- A valid certificate for HTTPS
- Tyk Gateway and Dashboard
- Set `flush_interval = 1 ms` in `tyk.conf`

Tyk API Configuration

1. Protocol: **HTTPS**
 2. Upload SSL certificate
 3. Custom domain: `tyk.com`
 4. Listen path: `/routeguide.RouteGuide`
RouteChat method will be used (bidirectional)
1. Target URL: `h2c://localhost:10000`

Testing Bidirectional Streaming

- Point client to `https://tyk.com:8000`
- Use the `RouteChat` service
- Client sends and receives data simultaneously
- Traffic:
 - External = Encrypted
 - Internal = Plain H2C